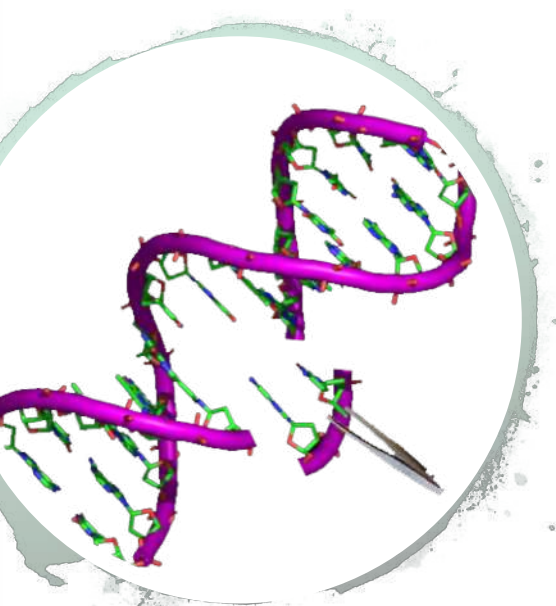




Molecular Techniques for Investigating Genetic Diseases

2nd year Medicine

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Objectives

- Define genetic testing.
- List indications of genetic testing.
- Explain different molecular biology techniques of genetic testing for different genetic disorders.
- Discuss the applications of genetic testing.

Important Definitions:

- **Pleiotropy:** A gene controls many phenotypic characteristics
- **Allelic Diseases:** Different diseases due to different mutations in the same gene
- **Genetic Heterogeneity:** A disease which has different genetic causes
 1. Allelic Heterogeneity: A disease which result because of different mutations in the same gene
 2. Locus Heterogeneity: A disease which result due to mutations in different genes
- **Polygeny:** The expression of a certain clinical characteristic is determined by several genes
- **Penetrance:** The likelihood that a genetic variant will trigger any symptom
- **Expressivity:** Type or extent of expression of the clinical features

Genetic testing

- A branch of medical investigations that involves analysis of individual's blood, skin, hair, or other body tissue to look at [chromosomes](#), [genes](#), and [proteins](#) for a change, or mutation, that is associated with a genetic condition.
- For genetic testing [we need cells that have a nucleus](#) (ERYTHROCYTES HAVE NO NUCLEUS).

Indications for Genetic Testing

- 1- Chromosomal disorders
- 2- Monogenic disorders



Genetic testing

The information gained from genetic testing may be helpful in a number of ways such as:

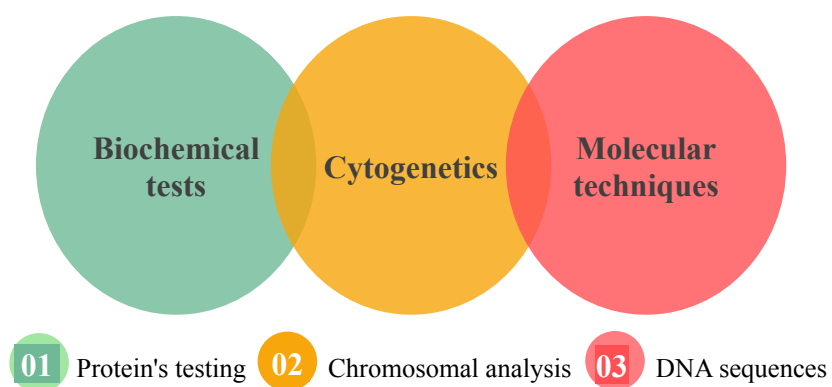
- diagnosing a genetic disease,
- starting treatment,
- initiating prevention strategies,
- Testing other healthy family members (**predictive test**) to control a genetic disease by Variant/mutation carriers as well as prevent the occurrence of the complications.
- as well as making life decisions such as career choice and family planning.

Box 5.1**Applications of DNA technology**

Gene structure/mapping/function
 Population genetics
 Clinical genetics
 Preimplantation genetic diagnosis
 Prenatal diagnosis
 Presymptomatic diagnosis
 Carrier detection
 Diagnosis and pathogenesis of disease
 Genetic
 Acquired — infective, malignant
 Biosynthesis (e.g., insulin, growth hormone,
 interferon, immunization)
 Treatment of genetic disease
 Gene therapy
 Agriculture (e.g., nitrogen fixation)

Table 5.1

Methods of Genetic Testing



What is molecular genetics?

Molecular genetics is the study of the molecular structure of DNA, its cellular activities (including its replication), and its influence in determining the overall makeup of an organism.

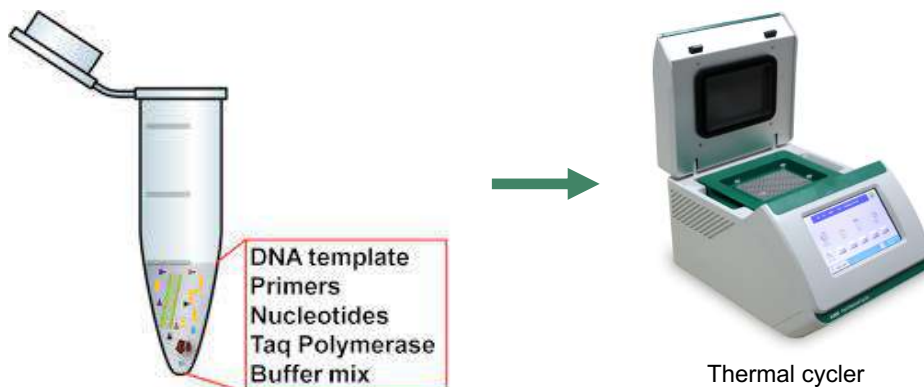
. Polymerase Chain Reaction (PCR)

- The polymerase chain reaction is an extremely versatile technique for copying DNA.
- PCR allows a single DNA sequence to be copied (millions of times), or altered in predetermined ways.
- Used in the early stages of processing DNA for sequencing .



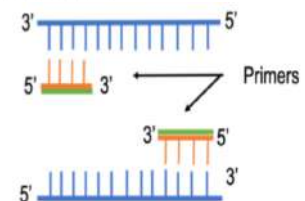
. Polymerase Chain Reaction (PCR)

Amplifying DNA *in vitro* by PCR= Amplimers

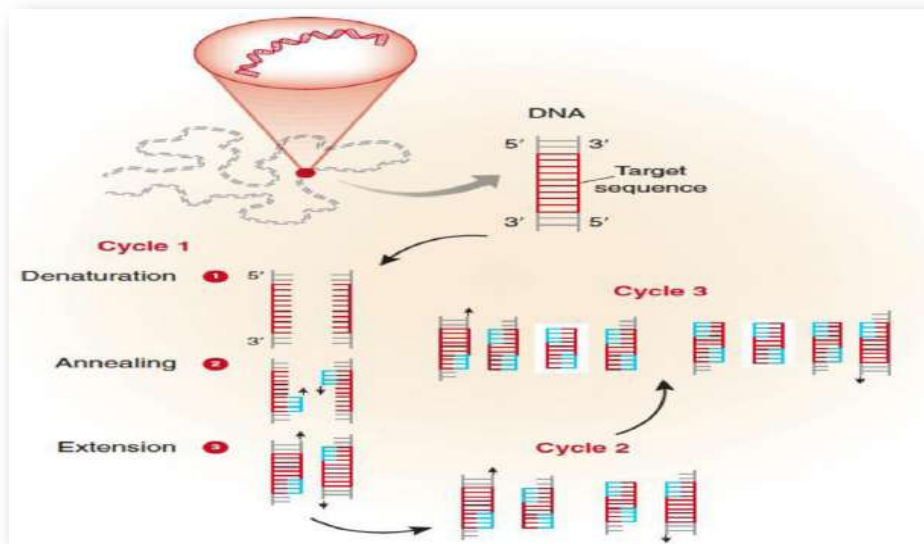


PRIMER

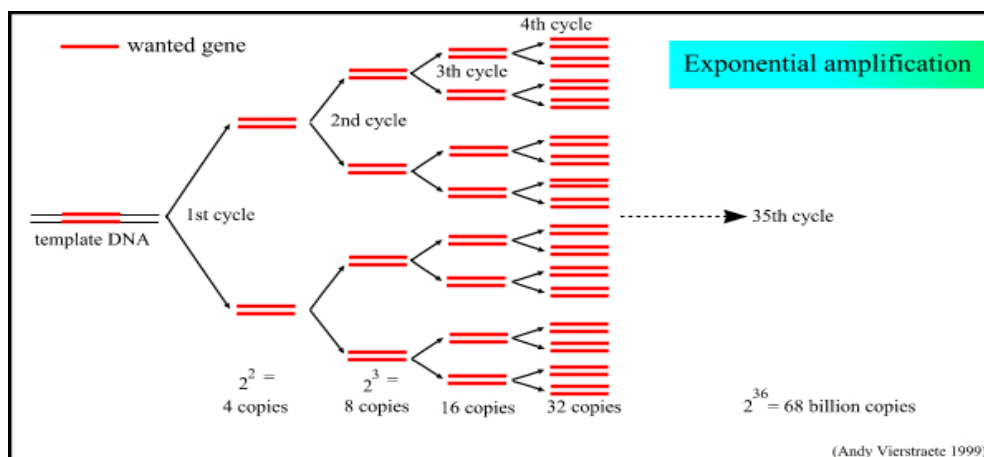
- A **primer** is a strand of nucleic acid that serves as a starting point for DNA synthesis.
- These primers are usually short, chemically synthesized oligonucleotides, with a length of about twenty bases. They are hybridized to a target DNA, which is then copied by the polymerase.
- Minimum primer length used in most applications is 18 nucleotides.
- Replication starts at the 3'-end of the primer, and copies the opposite strand.
- In most cases of natural DNA replication, the primer for DNA synthesis and replication is a short strand of RNA .



PCR Method



. Polymerase Chain Reaction (PCR)



The process follows the principle of DNA replication

Nucleic Acid Hybridization

Many methods of DNA analysis involve the use of nucleic acid probes and the process of nucleic acid hybridization.

Nucleic Acid Probes:

are usually single-stranded DNA sequences that have been radioactively or non-radioactively labeled and can be used to detect DNA or RNA fragments with sequence homology.

Nucleic Acid Hybridization:

involves mixing DNA from two sources that have been denatured by heat or alkali to make them single-stranded and then, under the appropriate conditions, allowing complementary base pairing of homologous sequences to identify a specific DNA sequences

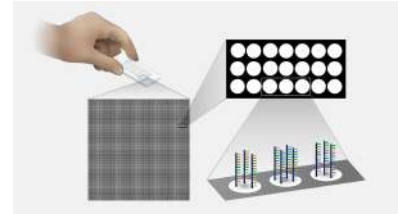
Molecular Methods

- **Dosage Analysis:**
 - Microarray
 - MLPA
- **Sequencing Methods:**
 - Sanger Sequencing
 - WES/WGS

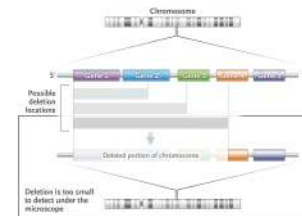
DNA-Array (Chromosomal Microarray, CMA)

Definition:

It is a **quantitative, molecular cytogenetic technique** that help to detect a genome-wide small chromosome changes (**< 5Mb, i.e.**, microdeletion or microduplication) which can not be detected with conventional chromosome analysis; (**microdeletion- microduplication syndromes**)



CMA has a resolution of around **1 Kb**, which is the **highest resolution** among current cytogenomic methods



<https://www.genome.gov/genetics-glossary/>

Indications of Microarray

Indications: To diagnose the cases of

1. **Developmental delay (DD)**
2. **Intellectual disability (ID)**
3. **Congenital anomalies and dysmorphic features in particular with normal karyotype**
 - With classical (conventional) chromosome analysis around **7-10%** of DD/ID cases can be diagnosed
 - with using CMA, **15-20%** of relevant chromosomal aberrations are detected

Limitations of Microarray

No evidence of **balanced rearrangement**
(Ex: balanced translocation and inversion)

It doesn't give information about the **position of the duplication**

Cannot detect **tiny duplications and deletions of DNA** segments within a single gene, single- nucleotide variants (SNVs)

It doesn't give information about **point mutation**

Limited evidence of **mosaic** (cannot detect mosaicism < 20—25%)

Examples of Microdeletion Syndromes: **22q11.2 Deletion Syndrome**

- Also known as **DiGeorge or Velocardiofacial syndrome**
- The molecular defect is a **3-Mb microdeletion on chromosome 22 (22q11.2)**
- Mode of inheritance: AD (90% de novo, 10% inherited from one of the parents)



Multiplex Ligation-Dependent Probe Amplification (MLPA)

This is a high-resolution method used to detect deletions and duplications on an exon level.

- It is the preferred method of analysis for the detection of large genomic deletions or duplications (>50 bp).
- It is a variation of the multiplex PCR that permits amplification of multiple targets with only a single primer

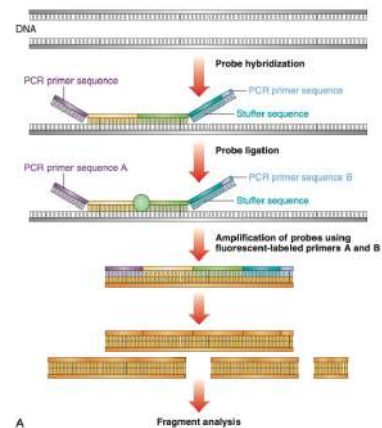
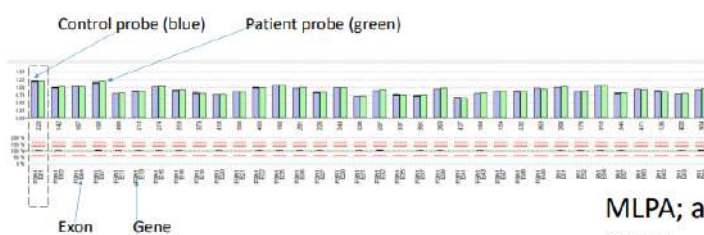


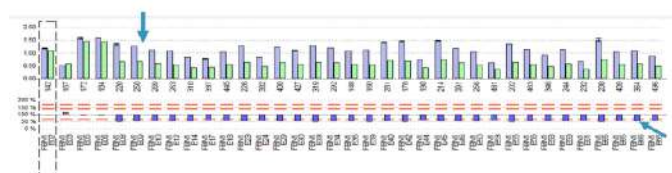
FIGURE 5.17 A, Illustration of multiplex ligation-dependent probe amplification (MLPA) method. Emery's Elements of Medical Genetics 15th Edition

Multiplex Ligation-Dependent Probe Amplification (MLPA)

Examples of MLPA; Evaluation (Normal)



MLPA; a heterozygous deletion of multiple exons



See the difference between the control (blue) and patient probe (green)

Example of disease which can be diagnosed with MLPA:

Duchenne Muscular Dystrophy (DMD)

- The most common and most severe form of muscular dystrophy

Responsible gene: *DMD* (Dystrophin)

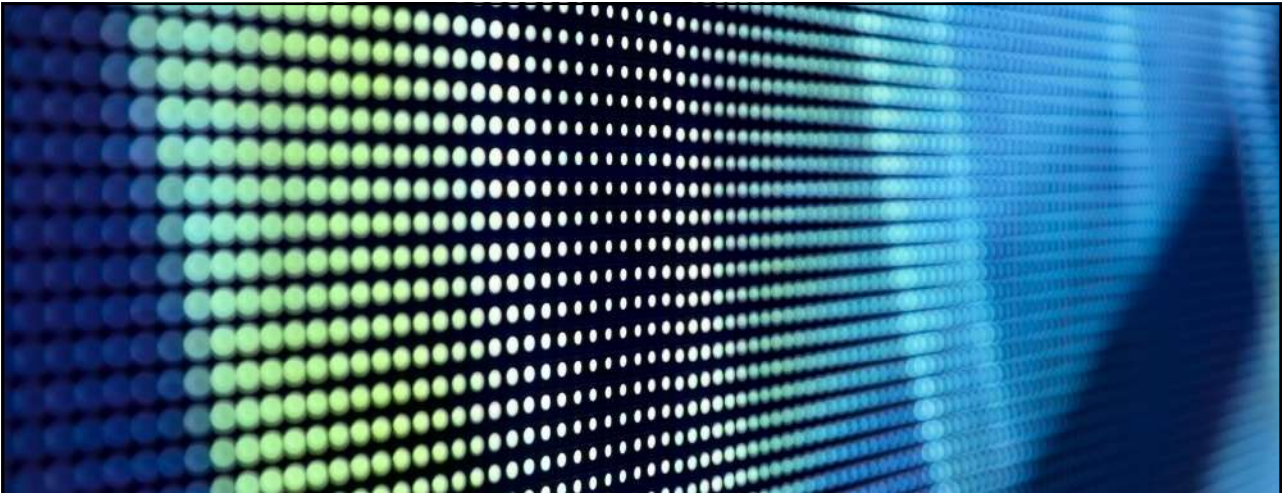
Cytogenetic locus: Xp21.2

Inheritance: X-linked recessive



At a Glance: Comparison of MLPA and Microarray

Feature 	MLPA	Microarray (CMA)
Scope	Targeted (known syndromes)	Genome-wide (broad scan)
Resolution	Very High (Exon-level)	High (usually 10-100 kb)
Coverage	40-50 specific regions	Whole Genome
Turnaround Time	Fast (1-2 days)	Slower (1-2 weeks)
Cost	Relatively low	Higher
Best Use	Validating Array/Known Syndrome	Unexplained ID/DD/Autism



Sequencing Methods

- Sanger Sequencing
- WES/WGS

Sequencing-Based Methods



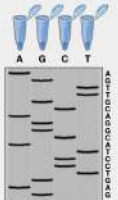
DNA sequencing by synthesis

Polynucleotide-based DNA sequencing

Sanger DNA sequencing

Sequence 500 - 700 DNA bases per reaction
18 reactions per gel

10 reactions per gel

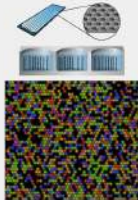


Sequenced 10,000 DNA bases per gel

Massively parallel DNA sequencing

Sequence 100 - 5,000 CpA bases per reaction
10 (forward to 10 billion reactions per site)

10 (forward) to 10 billion reactions per side



Sequence 2 trillion DNA bases per slide

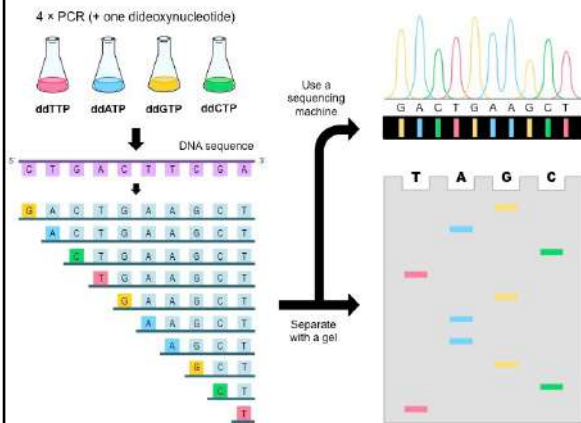
<https://www.genome.gov/genetics-glossary/>

Sequencing methods are the most common frequently used technique for mutation screening by suspecting of having a mutation within a specific gene or genes

Note:

The disease could be result of a mutation, two or several mutations in the same gene or several genes

Sanger Sequencing



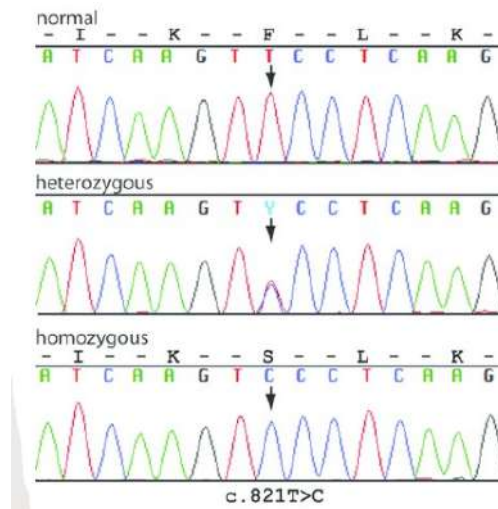
- Sanger was recently the **gold standard in mutation detection**
- Based on selective “chain termination” reactions occurring via in vitro DNA replication by DNA polymerase
- However, in genetic diagnostics it has increasingly been replaced by massively parallel sequencing processes (NGS)

<https://ib.bioninja.com.au/higher-level/topic-7-nucleic-acids/7.1-dna-structure-and-replication/dna-sequencing.html>

Sanger Sequencing Uses

1. It is still the method of choice for analysis a **specific mutation in a specific gene**
2. Analysis **small genes** with not more than around 27 exons; for example *CFTR* gene
3. Screening a **familial mutation** in the family members
4. Confirmation a detected mutation in NGS
5. Confirmation a *de novo* mutation in a patient (by testing the mutation in the parent)

Sanger Sequencing



Examples of diseases which can be diagnosed with Sanger Sequencing: **Cystic Fibrosis (CF)**

Responsible gene: *CFTR* (cystic fibrosis transmembrane conductance regulator)

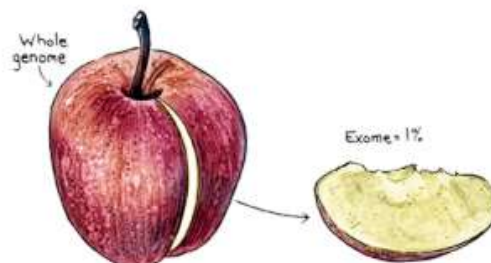
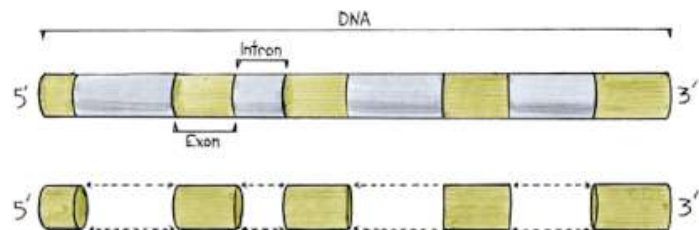
Cytogenetic locus: 7q31.2

Mode of inheritance: AR (one of the most common autosomal recessive disorders)

Next-Generation Sequencing (NGS)



Whole Exome Sequencing (WES) / Whole Genome Sequencing (WGS)



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Whole Exome Sequencing (WES)

- Method: Next generation sequencing
- Is a molecular technique the sequencing all protein coding regions in the genome (exons)
- Focuses on identifying disease-causing DNA variants within the 1.5-2% of the human genome

Whole Exome Sequencing (WES)

S L M H										P Q R S T U V W												
1	ref	alt	pos	ref	alt	pos	ref	alt	pos	ref	alt	pos	ref	alt	pos	ref	alt	pos	ref	alt		
44	chr23	25511376	25511376	G	T	het	255	540	57	55V	0.4549	exon1	SACS	nonstop	SACS_NM_014362:exon1:c.557G>A	p.L126V	0.004	0.997	0.0457	0.0052	694450	passed
154	chr6	16327969	16327970	TGC	ATG	het	217	882	44	18DEL	0.0034	exon1	ATXN6	nonstop	ATXN6_NM_000332:exon1:c.677G>A	p.L224D	0.997	12.000	0.0041	passed	694450	passed
162	chr6	16327969	16327969	G	A	het	215	1385	37	54V	0.5046	exon1	SYN1	stop	SYN1_NM_182943:exon1:c.224G>A	p.Q680X	0.997	10.000	0.0041	passed	694450	passed
163	chr6	16327969	16327969	C	T	het	215	697	37	54V	0.3626	exon1	SYN1	nonstop	SYN1_NM_182943:exon1:c.224G>A	p.Q680X	0.997	10.000	0.0041	passed	694450	passed
165	chr6	16327969	16327969	GA	AA	het	217	1353	37	18DEL	0.0027	exon1	SYN1	frameshift	SYN1_NM_182943:exon1:c.1587G>del	p.S529L	0.976	0.9409	0.0055	passed	694450	passed
204	SYN1_NM_182943:exon1:c.224G>A p.Q680X, valid; VUS; 23.04.2013 PB valid; 01.07.2013 JB																					
205	SYN1_NM_182943:exon2:c.1587G>del, valid; VUS; 23.04.2013 PB valid; 01.07.2013 JB																					
206																						
1	ref	alt	pos	ref	alt	pos	ref	alt	pos	ref	alt	pos	ref	alt	pos	ref	alt	pos	ref	alt		
44	0.4549	exon1	SACS	nonstop	SACS_NM_014362:exon1:c.557G>A	p.L126V	0.004	0.997	0.0457	0.0052	694450	passed	0.0000	0.0000	10.000	0						
154	0.0034	exon1	ATXN6	nonstop	ATXN6_NM_000332:exon1:c.677G>A	p.L224D	0.997	12.000	0.0041	passed	694450	passed	0.0000	0.0000	10.000	0						
162	0.5046	exon1	SYN1	stop	SYN1_NM_182943:exon1:c.224G>A	p.Q680X	0.997	10.000	0.0041	passed	694450	passed	0.0000	0.0000	10.000	0						
163	0.3626	exon1	SYN1	nonstop	SYN1_NM_182943:exon1:c.224G>A	p.Q680X	0.997	10.000	0.0041	passed	694450	passed	0.0000	0.0000	10.000	0						
165	0.0027	exon1	SYN1	frameshift	SYN1_NM_182943:exon1:c.1587G>del	p.S529L	0.976	0.9409	0.0055	passed	694450	passed	0.0000	0.0000	10.000	0						
204	SYN1_NM_182943:exon1:c.224G>A p.Q680X, valid; VUS; 23.04.2013 PB valid; 01.07.2013 JB																					
205	SYN1_NM_182943:exon2:c.1587G>del, valid; VUS; 23.04.2013 PB valid; 01.07.2013 JB																					
206																						

Example of evaluation and analysis with NGS technique

Indications of WES

- cases of suspected hereditary diseases which cannot be clearly assigned to a genetic syndrome or even narrowed down to potential mutations based on clinical features
- So, **to diagnose genetic disorders, that result from mutation in the gene**
- Analysis of the clinical exome involving the healthy parents of an affected patient (**trio analysis**) can, in particular, also be a good possibility to uncover a suspected de-novo mutation
- Solo....
- Duo.....

Advantages of WES

- Very **rare genetic causes of** disease can be identified
- **Identify novel genes** underlying known or unrecognized genetic disorders
- Allowed the detection of “**blended phenotypes**” (two or more genetic diagnoses in a single individual, resulting in mixed clinical features)
- Can lead to the **phenotypic expansion of** previously described genetic syndromes, due to the detection of mildly affected patients

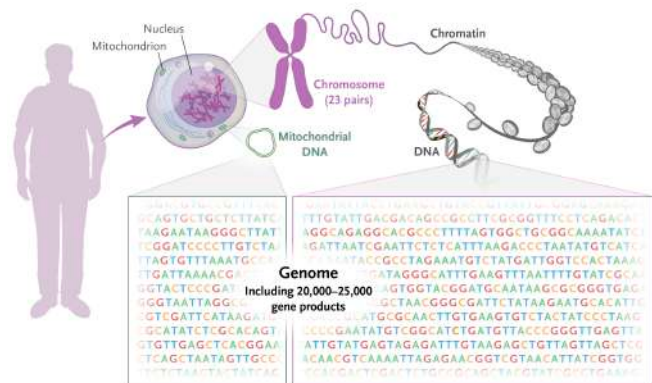
Whole Genome Sequencing (WGS)

The human genome includes ALL coding and noncoding nuclear and mitochondrial DNA sequences.

- WGS include sequencing and analyzing an individual's entire genome, coding and noncoding regions (i.e., exons + introns)

Whole Genome Sequencing (WGS)

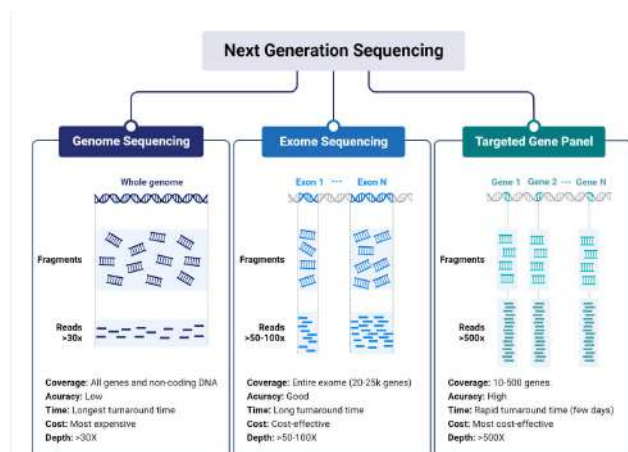
- WGS is also performed via NGS but does not require the targeted sequence capture step, as in WES and targeted panels
- Able to detect CNVs smaller than three exons, and it could potentially identify balanced chromosomal rearrangements
- Also analyzes the mitochondrial DNA
- Analysis of the WGS data can reveal triplet repeat expansion
- The major challenge is data storage and interpretation of genome sequence data



<https://illustrated-glossary.nejm.org/>

Difference Between Whole Exome Sequencing (WES), Whole Genome Sequencing (WGS) and Gene Panel

- **WES** is focused on protein coding regions (exon; ~1.5–2% of the genome)
- **WGS** is a comprehensive test that covers up to 98% of the whole human genome (exons and introns)
- **Gene Panel** is a test that analyses multiple genes which related with a specific genetic disorders;
- Ex: hereditary breast and ovarian cancer panel (HBOC panel)



What is the Difference between:

Conventional chromosomal analysis	Chromosomal changes (>5Mb)
FISH-analysis	Specific chromosomal changes
Microarray	Microdeletion/duplication (<5Mb)
MLPA	Gene exon-level deletion / duplication
Sanger Sequencing	Gene mutation in a gene – point mutation
Gene panel	Group of genes
WES	Look for a mutation in exons
WGS	Look for a mutation in exons and introns



Thank you